**Task 5: Transaction-Based Payment Simulation**
Description: Simulate an online payment process,Deduct balance from user account ,Add amount to merchant account ,Use COMMIT on success, ROLLBACK on failure ,Real-Time Usage: Banking and digital payment applications.

1. Objective of the Task
The goal is to simulate how an online payment system works using database transactions. In real systems like banking apps, money transfer is done safely using COMMIT and ROLLBACK to ensure data consistency.

2. Key Idea
A payment process involves two main operations:
- Deducting money from the user account
- Adding money to the merchant account

Both operations must succeed together. If any one fails, the whole transaction must be undone.

3. Database Concept Used
We use:
- COMMIT → Saves all changes permanently if everything is successful
- ROLLBACK → Cancels all changes if something goes wrong

This ensures no money is lost or duplicated.

Transaction Simulation (Correct Way)
```
START TRANSACTION;

-- Step 1: Deduct from user
UPDATE users
SET balance = balance - 500
WHERE user_id = 1;

-- Step 2: Add to merchant
UPDATE merchants
SET balance = balance + 500
WHERE merchant_id = 1;

-- Step 3: If everything is OK
COMMIT;

ROLLBACK;

SELECT * FROM users;
```

OUTPUT:

```
mysql>     COMMIT;
Query OK, 0 rows affected (0.001 sec)

mysql> ELSE
    ->     ROLLBACK;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'ELSE
    ROLLBACK' at line 1
mysql>     SELECT 'Transaction failed: Insufficient balance' AS Message;
+-----------------------------------------+
| Message                                 |
+-----------------------------------------+
| Transaction failed: Insufficient balance |
+-----------------------------------------+
1 row in set (0.000 sec)

mysql> END IF;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'END IF' at line 1
mysql> SELECT * FROM users;
+---------+---------+---------+
| user_id | name    | balance |
+---------+---------+---------+
|       1 | Krishna | 4000.00 |
|       2 | Arun    | 3000.00 |
+---------+---------+---------+
2 rows in set (0.001 sec)

mysql> SELECT * FROM merchants;
+-------------+---------------+----------+
| merchant_id | name          | balance  |
+-------------+---------------+----------+
|           1 | Online Store  | 17000.00 |
|           2 | Food Delivery |  5000.00 |
+-------------+---------------+----------+
2 rows in set (0.000 sec)

mysql>
```

3.Conclusion

- This simulation shows how transaction-based payment systems work in databases. The system ensures:
- Money is transferred safely between user and merchant accounts
- COMMIT is used to save successful transactions
- ROLLBACK is used to undo failed transactions
- Errors occurred because IF-ELSE cannot be used directly in MySQL terminal